

enzymes proteins that control a reaction in a cell

genome the total genetic material in a cell or organism

polymers molecules composed of many similar parts

protein need to be sequenced. That information can then be entered into a database and matched with known proteins having identical or related sequences.

Sequencing a protein from its C-terminus is particularly challenging, and there are no techniques that are as robust as Edman degradation. However, some limited amino acid sequence information can be obtained using **enzymes** called carboxypeptidases, which remove individual C-terminal amino acids. These enzymes, however, tend to cleave only specific amino acids from the C-terminus.

Carboxypeptidase B, isolated from cow pancreas, for example, can release the amino acids arginine and lysine from the C-terminus of a protein. Carboxypeptidase A, also isolated from cow pancreas, fails to release arginine, lysine, or proline, but can cleave off the other seventeen amino acids. Carboxypeptidases isolated from citrus leaves and yeast can cleave off any amino acid from the C-terminus of a protein, although the rate at which they do this depends on the particular amino acid. If one amino acid is released slowly and the next within the chain is released very quickly, they might appear to be cleaved at the same time, making it difficult to establish their order. C-terminal amino acid identification using enzymes, therefore, is not practical beyond the first several positions.

Another method of protein sequencing, called mass spectrometry, uses electric current to break individual amino acids from a protein. In a mass spectrometer, the released amino acids are collected in a detector and are each identified by their unique mass.

Sequencing of the human **genome** has allowed a giant leap in the understanding of how the human species evolved and how genetic diseases arise. Advances made in DNA sequencing technology lead to this grand accomplishment. The next frontier is to decipher how all the proteins encoded by the genome interact to carry out the processes of life. This is the study of proteomics. Advances in mass spectrometry and protein sequencing instrumentation are bringing this challenging problem closer to its resolution. SEE ALSO HPLC: HIGH-PERFORMANCE LIQUID CHROMATOGRAPHY; MASS SPECTROMETRY; PROTEINS; SEQUENCING DNA.

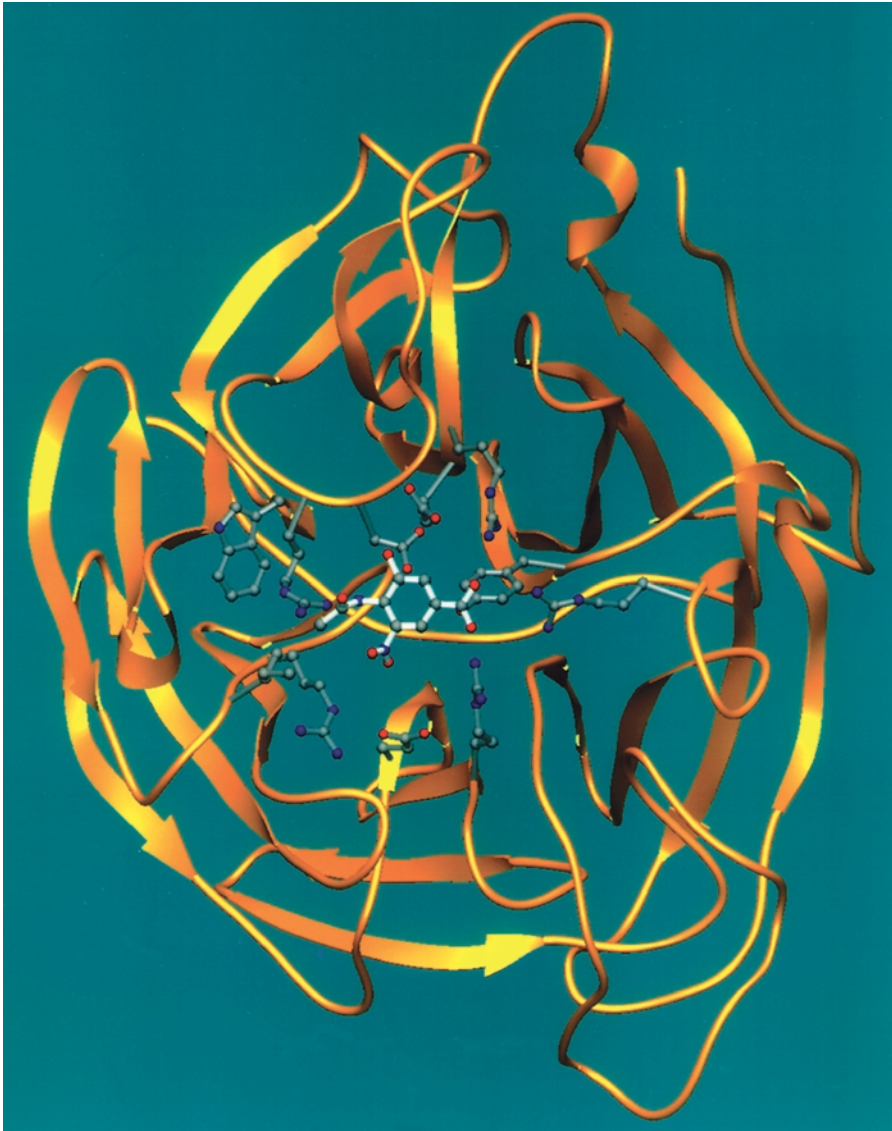
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Proteins

Proteins are **polymers** of amino acids that provide structure and control reactions in all cells. When humans think of expressing the meaning of life, they often resort to words. From poems to sonnets to short stories to novels, words tell the stories of life. But in biological terms, the words of life are proteins. While DNA holds the code of life, proteins are the language in which that code is expressed.



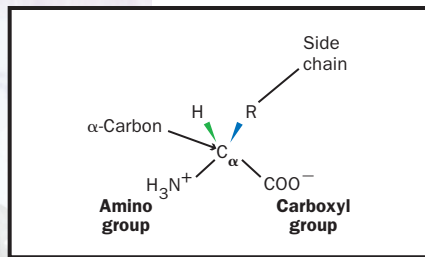
Computer generated representation of the enzyme neuraminidase (ribbons and strings) with an inhibitor (ball and stick) in the active site, a cleft on the enzyme's surface. This tertiary structure is built up from the primary structure (amino acid structure) and secondary structure. The arrows represent the beta-pleated sheet, a type of secondary structure.

To observe the mosaic of proteins in life is to observe nature in its finest array. The feathers of a bird and the silk of a spider's web are both almost pure protein. The most numerous proteins in an animal are the collagen proteins joining animal body parts. Other proteins include the positively charged histone proteins that condense the cell's negatively charged DNA and the transcription factor proteins that control which genes are expressed (made into proteins) and which remain silent. A plant traps CO_2 to make sugar with Earth's most abundant protein, the **enzyme** ribulose 1,5-biphosphate carboxylase. The protein hemoglobin transports gases through the bloodstream necessary for the metabolism of life. Other proteins store minerals (ferritin) or fats (ovalbumin), contract muscles (myosin), protect against infection (antibodies), or act as toxins (botulinum) or hormones (insulin).

Properties of Amino Acids

The English language consists of thousands of words, created from any of twenty-six letters arranged in a precise order. In an analogous fashion,

enzyme a protein that controls a reaction in a cell



Amino acid structure.
Adapted from Robinson,
2001.

hydrophilic “water-loving”

polar partially charged, and usually soluble in water

hydrophobic “water hating,” such as oils

nonpolar without charge separation; not soluble in water

basic having the properties of a base; opposite of acidic

acidic having the properties of an acid; the opposite of basic

codons sequences of three mRNA nucleotides coding for one amino acid

hydrogen bonding weak bonding between the hydrogen of one molecule or group and a nitrogen or oxygen of another

proteins are made up of twenty common amino acids in a precise order dictating the protein’s structure and function. Every amino acid has a common structure, in which a central carbon is covalently bonded to a carboxyl group (COOH), an amino group (NH₂), a hydrogen, and a variable “R” group.

The chemical properties of the R group are what give an amino acid its character. The R group can be **hydrophilic** (attracted to water and other **polar** molecules) or **hydrophobic** (attracted to **nonpolar** molecules and repelled by water or other polar molecules). Hydrophilic R groups can have **basic** charges, as in the amino acid valine, or **acidic**, as in glutamic acid, or they may even be an uncharged polar group such as –OH (alcohol) or –NH₂ (amino), as in serine. A nonpolar or hydrophobic R group can be a hydrocarbon chain, as in leucine. There are also three special amino acids: cysteine, glycine, and proline. Cysteine has a reactive sulfhydryl R group that forms disulfide bridges (S–S) between regions of the protein chain. These bridges increase toughness and resistance to unfolding of the protein structure. Glycine is the smallest amino acid, with hydrogen as its R group, and it fits into tight places within a protein’s structure. Proline has a cyclic ring involving the central carbon, and it causes kinks to occur in a protein chain. Both proline and glycine are common at the corner of turns in the protein foldings.

Primary Structure

The unique sequence of amino acids in a protein is termed the primary structure. When amino acids form a protein chain, a unique bond, termed the peptide bond, exists between two amino acids. The sequence of a protein begins with the amino of the first amino acid and continues to the carboxyl end of the last amino acid.

The unique sequence of amino acids results from the translation of **codons** present in messenger RNA (mRNA). The mRNA, in turn, is a complementary copy of the gene that codes for that protein. Protein structure and function can change when “misspellings” occur in the order of amino acids during their transcription and translation. Sickle-cell hemoglobin, for example, is “misspelled” in only one amino acid; the sixth amino acid in the beta chain, where a valine is substituted for a glutamic acid. This occurs because the codon for valine, GUG, has replaced the codon for glutamic acid, GAG. This change from acidic to basic amino acid causes the hemoglobin molecules to stick to one another, forming long chains and blocking oxygen binding. These chains of hemoglobin precipitate in the cell, causing the red blood cells to assume a sickle shape. All of these structural and functional changes occur because of the mutation in the hemoglobin gene and a “misspelling” in the hemoglobin’s amino acid sequence.

Secondary Structure and Motifs

The secondary structure of proteins is due to foldings that occur within their structure. These foldings are either in a helical shape, called the “alpha-helix” (which was first proposed by Linus Pauling), or a beta-pleated sheet shaped similar to the zig-zag foldings of an accordion. The turns of the alpha-helix are stabilized by **hydrogen bonding** between every fourth amino acid in the chain. The alpha-helix can cover specific regions of the protein

or it may involve the entire protein, as in the alpha-keratin found in claws and horns. The two sides of the alpha-helix may differ in polarity, with hydrophilic R groups projecting to the lining of the channel, while hydrophobic R groups project to the outside of the channel, where they embed in the hydrophobic membrane. This structure is exemplified in membrane channel proteins, proteins that channel ions across from one surface to another. The beta-pleated sheet is formed by folding successive planes. Each plane is five to eight amino acids long. The folds are stabilized by hydrogen bonding. The strength observed in silk fibers is due to their stacks of beta-pleated sheets.

Combinations of secondary structure form “motifs.” A coiled-coil motif is common among proteins that associate with the DNA helix. The helix-loop-helix motif is a knobby structure, and the zinc finger projects outward like its name. These last two motifs allow associations between RNA and proteins that form the basis of their interactions.

Tertiary Structure and Protein Domains

Domains are large functional regions of the protein, such as an enzyme’s active site, which binds the substrate to the enzyme. Myoglobin, the muscle protein that stores and releases oxygen, contains several alpha-helices wound around a central crevice. It is in this central crevice that the O_2 molecule binds. Just as words take on their meanings when completed, the functional domains unite to form the overall purpose of a protein. For example, a membrane protein stabilizes itself by anchoring itself with a hydrophilic **cytoplasmic** domain, then weaves its alpha-helices throughout the membrane domain and projects its carbohydrate hydrophilic side chains into the extracellular surface domain. Such membrane proteins often act as receptors, important for receiving signals such as hormones, or work in the immune system to recognize infected cells.

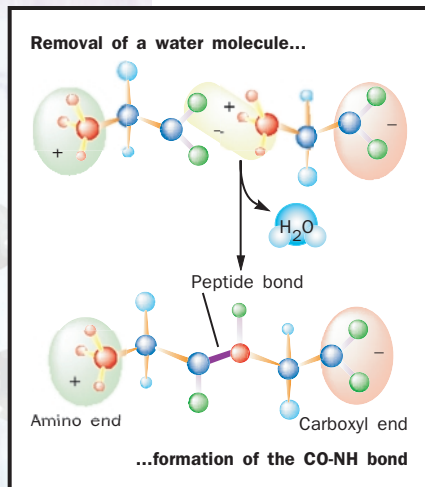
The local foldings, evident in secondary structure, then combine into a single polypeptide chain. This chain is called the tertiary structure, or **conformation**. For example, the pancreatic enzyme ribonuclease, which aids in digestion of RNA in the diet, consists mainly of beta sheet folds, with three small alpha-helical regions. Tertiary structure is often stabilized by disulfide bonds between adjacent cysteine in different regions of the protein. For example, the tertiary structure of ribonuclease contains four disulfide bonds, located at specific sites. The stability of the tertiary structure of proteins is destroyed by toxic heavy metals such as mercury. Concentrations of mercury in the environment, for example, result in the displacement of hydrogen on the sulfur atom (SH), thereby blocking functional disulfide bonds.

Several other weak, noncovalent interactions also help stabilize tertiary structure. These noncovalent interactions can be disrupted by heating a protein or exposing it to extremes in pH (acidity or alkalinity), which alters the charge of polar groups on the amino acids. Such disruptions cause the protein to unfold, often exposing hydrophobic groups and leading to precipitation (clumping together) of the protein. If these disruptive factors are removed, some proteins can refold to their original conformation. This ability to refold confirms that protein folding is a self-assembly process that is dependent upon the sequence of amino acids.

cytoplasmic describes the material in a cell, excluding the nucleus

conformation three-dimensional shape





Peptide bond formation.
Adapted from Robinson,
2001.

multimeric describes a structure composed of many parts

antigen a foreign substance that provokes an immune response

heme iron-containing nitrogenous compound found in hemoglobin

plasma membrane
outer membrane of the cell

endoplasmic reticulum
network of membranes within the cell

lumen the space within the tubes of the endoplasmic reticulum

Quaternary Structure

Some proteins need to functionally associate with others as subunits in a **multimeric** structure. This is called the quaternary structure of the protein. This can also be stabilized by disulfide bonds and by noncovalent interactions with reacting substrates or cofactors. For example an antibody consists of two “light” polypeptide chains covalently linked to two longer “heavy” chains, forming a Y-shaped molecule with each branch able to bond to an identical **antigen**. The protein subunits of the single-stranded binding protein of *Escherichia coli* bind to DNA only as a tetramer (a multimeric form), acting to stabilize the separated DNA strands during replication.

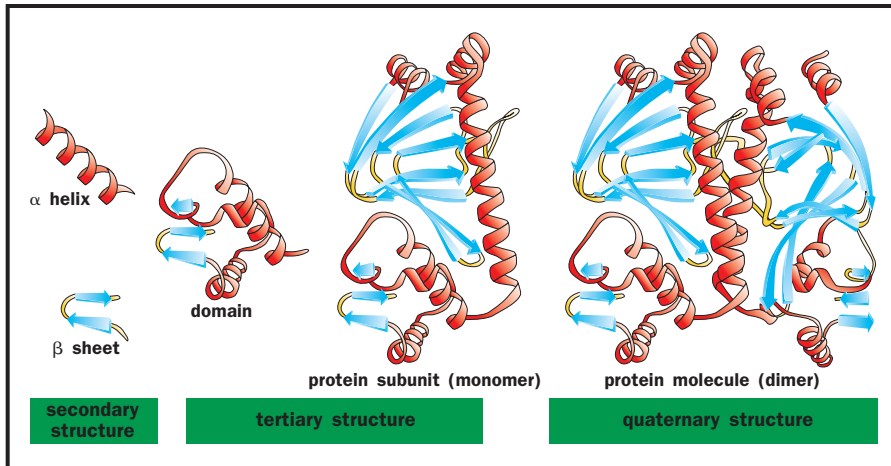
Another excellent example of quaternary structure is that of hemoglobin. Adult hemoglobin consists of two alpha subunits and two beta subunits, held together by noncovalent interactions. Each of the four subunits contains a **heme** group that binds an oxygen molecule, O_2 . This binding of oxygen is a cooperative process whereby the binding of one oxygen molecule occurs slowly, but once achieved then speeds the binding of the remaining three oxygen molecules. The fourth oxygen molecule binds 300 times faster than the first oxygen molecule. This cooperativity assures that maximum oxygen is captured and retained as it enters into the capillaries within the lungs.

The unloading of oxygen is also facilitated by cooperativity, such that after one oxygen molecule is released, the other three soon follow. This assures that the tissues will receive maximum oxygen once it is delivered. Alpha-hemoglobin by itself, or tetramers of all beta subunits, also bind oxygen, but not with the same cooperativity. Such evidence indicates that there is some form of molecular interaction between the subunits of the tetramer of adult hemoglobin.

Signal Sequences in Protein Synthesis

Protein must be delivered to the proper destination in the cell to function properly. Signal sequences within the protein itself act like “zip codes” to ensure correct delivery. The synthesis of secreted proteins like insulin and of proteins that will be integral to the **plasma membrane** occurs at a ribosome tethered to the **endoplasmic reticulum**, which is a system of membranes that transport materials within cells. The peptides formed there are then translocated into the **lumen**, or channel, of the endoplasmic reticulum, where they will be formed into a polypeptide chain. This translocation occurs because of a specific signal sequence that is formed by the first twenty or so amino acids in the protein. The core of this sequence consists of ten to fifteen amino acids that have hydrophobic side chains such as alanine, leucine, valine, isoleucine, and phenylalanine, which are usually cleaved from the protein later on. The nascent polypeptide chain is guided along this path by a signal receptor protein.

Proteins targeted for internal cellular functions are synthesized on ribosomal assemblages that float free in the cytoplasm. Such proteins also have their signal sequences. Proteins destined for the cell’s nucleus have a specific nuclear signal sequence consisting of a small series of basic amino acids such as arginine and lysine bounded by proline. This nuclear signaling sequence can be located anywhere in the protein’s sequence as long as it projects outward from the three-dimensional tertiary structure. Signal sequences for proteins targeted to be part of organelles such as the mito-



Protein structure components. Alpha helices and beta sheets are linked by less-structured loop regions to form domains. The domains combine to eventually form fully functional proteins. Adapted from Robinson, 2001.

chondria and chloroplasts are anywhere from twenty to seventy amino acids long and are mostly hydrophilic. This charged nature allows easy travel through the hydrophilic cytoplasm to the organelle.

Molecular Chaperones

Although the folding of the protein into its tertiary structure is determined by the primary order of amino acids, the process of folding occurs with the assistance of molecular chaperone proteins. These molecular chaperones often have pockets or tunnels that envelop the nascent polypeptide. This enveloping allows the folding of the protein to occur unhindered by unwanted interaction with other cellular components.

Chemical Modification and Processing of Proteins

Most proteins are structurally altered after synthesis through chemical modification or processing. These alterations help the cell determine a protein's fate, such as whether that protein is active or inactive, how long the protein will function, and to some degree the location where that protein will function. Chemical modifications, which are additions of chemical groups to the R groups in the amino acids, are made after translation. Such modifications may include the attachment of a phosphate group (phosphorylation) to the alcohol group on the amino acids of serine, threonine, or tyrosine. The amino acid proline in proteins such as collagen is often hydroxylated, which means that an alcohol group is attached. Other amino acids with amino groups in their R region, such as lysine or arginine, may be chemically modified through methylation, which is the addition of a methyl group ($-\text{CH}_3$), or through acetylation, in which an acetyl group ($-\text{CH}_3\text{CO}$) is added. Larger modifications, such as the addition of a carbohydrate group, occur to create **glycoproteins** in specialized organelles termed Golgi apparatus.

Modifications change the charge of the protein, and often cause a change in the protein's activity level. For many DNA-associated proteins their regional acetylations cause them to "loosen" their grip on the DNA helix, thereby enabling **transcription factors** to enter, signaling gene activation. A cascade of internal protein phosphorylation (successive additions of a phosphate group) is a common mechanism for carrying a hormone's message

glycoproteins proteins to which sugars are attached

transcription factors proteins that increase the rate of gene transcription

from the membrane, where it docks into the cell and induces a metabolic change inside the target cell.

Processing results in cutting off specific parts of the protein (cleavage). Many digestive proteins such as pepsin and hormones such as insulin are processed. Pepsin, which is a digestive protein secreted into the lumen of the stomach, remains in an inactive form until stomach acid is also secreted. The timing of the acid secretion, pepsin activation, and entry of food coincide so that pepsin's activity will be directed toward the food and not the wall of the stomach.

Conformational Changes in Protein Structure

As noted above, a protein's activity can be regulated when it undergoes a change in its conformation. A dramatic and extensively studied model of protein conformational change is that of the Na^+/K^+ ATPase pump. This is an integral membrane protein with one side facing the exterior of the cell and the other facing the cytosol. It is used for the specific transport of sodium or potassium across the membrane, and one of its most important functions is the repolarization of a nerve fiber after it "fires."

The first step in the transport process is the binding of three Na^+ (sodium) ions to the inside face of the protein. This is followed by protein phosphorylation using **ATP**, which causes the protein to change its conformation. This moves the sodium ions from the cytosol to the exterior. This conformational change also opens up exterior binding sites, which tightly bind two potassium ions outside the cell. Following the potassium binding, the protein is dephosphorylated, losing its recently added phosphate group. This dephosphorylation then changes the protein back to the original conformation, causing the protein to loosen its binding of potassium and deliver those two ions to the cytosol. This process demonstrates that protein structure can be reversibly changed. The net result is that the inside of the cell develops a slight negative charge compared to the outside. The disruption of this "polarized" state constitutes nerve cell firings, which allow the cells of the nervous system to communicate with one another.

ATP adenosine triphosphate, a high-energy compound used to power cell processes

Proteomics

Proteomics is a new field of study that seeks to describe which proteins are expressed in a cell, when they are expressed, what consequences result from their expression, and how they fit into biochemical pathways. The first step in the study of proteomics is to define the language of protein structure. The field of proteomics promises to bring a complex understanding to the role of proteins in living cells. **SEE ALSO** CELL, EUKARYOTIC; CHAPERONES; GENETIC CODE; HEMOGLOBINOPATHIES; IMMUNE SYSTEM GENETICS; MUTATION; NUCLEASES; PROTEOMICS.

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Proteomics

Proteomics is the science of studying the multitude of proteomes found in living organisms. A proteome is the entire collection of proteins expressed by a genome or in a tissue. The contents of a proteome can differ in various tissue types, and it can change as a result of aging, disease, drug treatment, or environmental effects.

This is contrary to the concept of a genome, which is an organism's complete collection of DNA. A genome's composition remains more or less constant from tissue to tissue, except for mutations and **polymorphisms** that can occur.

polymorphisms DNA
sequence variants

The word “proteome” was first coined in late 1994. By 1997 there were a number of research conferences focusing on proteomics.

According to the first draft of the human genome, based on the work by the Human Genome Project and by Celera Inc., there are only between thirty thousand and seventy thousand genes in the human genome, many fewer than had been estimated previously. However, as of 2002 there were still groups that believed that there are at least 120,000 genes. Regardless of which of these estimates proves more accurate, the number of potential proteins in the human proteome is quite large. Although the first draft of the human genome reduced the estimates for the total number of human genes, it also predicted a greater amount of alternative splicing of genes, and therefore more distinct protein products per gene, than had been anticipated.

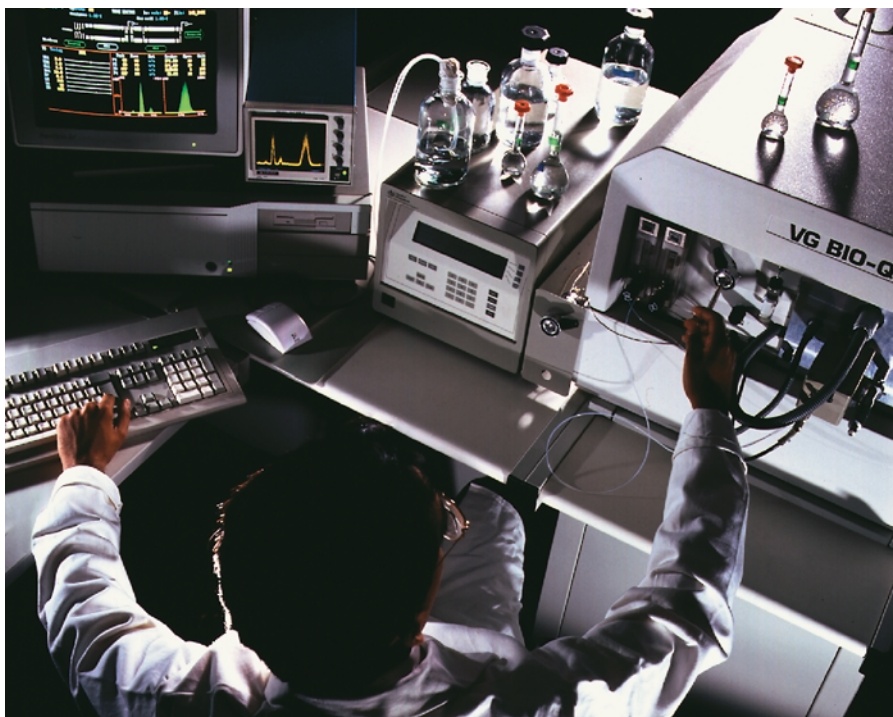
At its simplest level, proteomics is the study of protein expression in a proteome, or trying to understand the relative levels (amounts) of each protein within the mixture. Proteomics attempts to characterize proteins, compare variations in their expression levels in normal and disease states, study their interactions with other proteins, and identify their functional roles.

Unlike the traditional approach of studying individual proteins one at a time, proteomics uses an automated, high-throughput approach. High-throughput refers to the number of items (in this case, proteins) that can be analyzed or studied per unit of time. New technologies and substantial bioinformatics tools are required to compare entire proteomes. Expansion of the field of proteomics into the realm of “big science” (meaning many dollars invested by a large number of companies and universities) is several years behind the expansion of genomics. This is primarily because proteins are more difficult to work with in a laboratory setting than are nucleic acids such as DNA.

The development of protein analysis technologies is more difficult than the development of DNA analysis technologies for three reasons. First, the basic alphabet for encoding proteins consists of twenty amino acids, whereas there are only four different nucleotides, the alphabet of DNA. Second, the messenger RNA (mRNA) for some genes can be differentially spliced, meaning that multiple messages can be made from a single gene, resulting in multiple, distinct protein products. Finally, many proteins are modified once they have been synthesized. This is known as post-translational modification. There are a number of types of post-translational modifications, such as the addition of sugar, phosphate, sulfate, lipid, acetyl, or methyl groups.



Mass spectrometry systems are used to help scientists analyze the various proteomes within an organism.



Each of these modifications has the ability to change the functional activity of a protein.

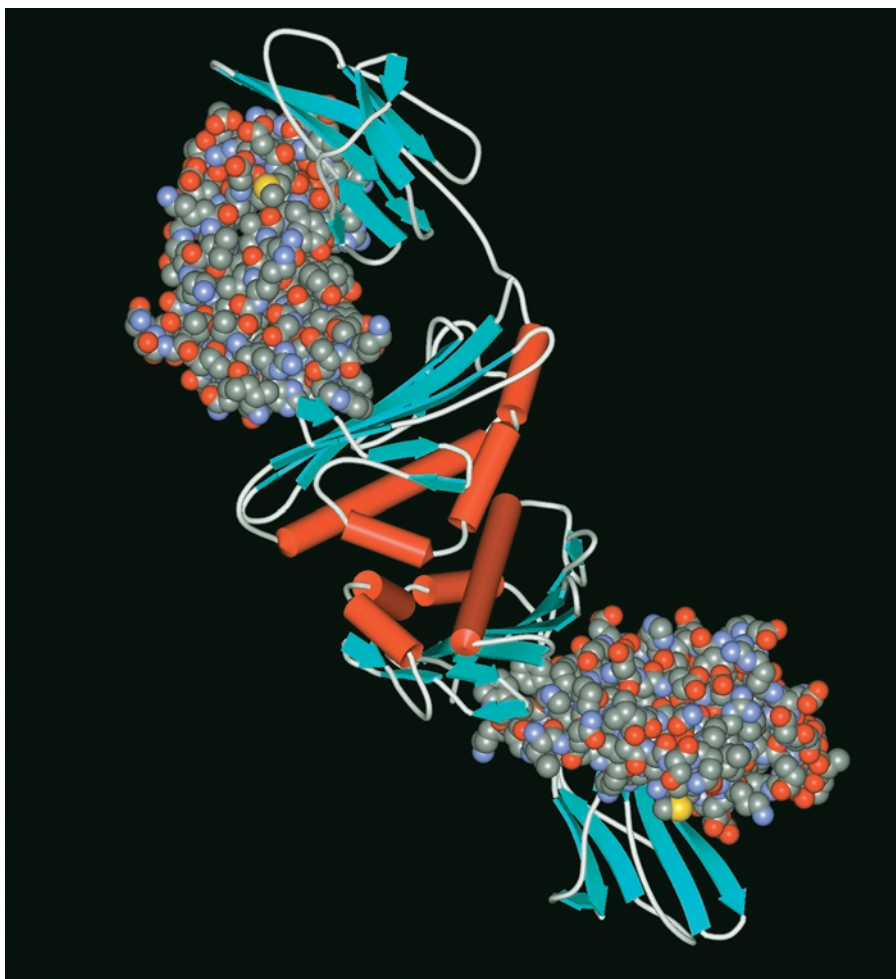
The above issues have made the elucidation of reliable, high-throughput techniques for characterizing proteins, including their expression levels, on a proteome-wide level a major challenge. Hence, techniques for doing, for example, high-throughput DNA sequencing and gene expression studies have been developed and commercialized on a large scale sooner than similar protein analysis techniques. This is not to imply that all of the techniques involved in proteomics are new. Some, such as two-dimensional **gel electrophoresis**, have been around since the 1970s. However, the need to adapt these techniques to a large “proteome” scale brings with it a unique set of challenges.

For researchers involved in areas such as drug discovery, proteomics approaches will need to be used to obtain a greater understanding of disease mechanisms and drugs’ mechanisms of action. Large-scale studies looking at gene expression via quantification of mRNA abundance are already possible and well commercialized. These technologies are very powerful, and the highest throughput approaches are capable of analyzing tens of thousands of genes per experiment. Sophisticated bioinformatics systems have been, and continue to be, developed to analyze these vast amounts of data. However, studies have shown that mRNA levels do not necessarily correlate well with protein levels.

Researchers must understand proteins and their roles, since proteins are the functional units within cells. As of 2002, the vast majority of drug targets were proteins. There are a handful of drugs, including some chemotherapeutic agents, that bind to DNA, but most drugs bind to specific protein targets. In the cases where the target is a protein, the drugs themselves are primarily small inorganic molecules or, in some cases, small proteins, such as **hormones**, that bind to a larger protein target in the body.

gel electrophoresis
technique for separation
of molecules based on
size and charge

hormones molecules
released by one cell to
influence another



Proteomics can help researchers understand how proteins interact in cells.

Some drugs are actually therapeutic proteins that are delivered to the site of the disease.

Laboratory Techniques

The primary attributes used to identify proteins include the protein's mass and apparent mass, its isoelectric point, and its N- and C-terminal sequence tags. A protein's mass and its apparent mass are probably the most common characteristics used. Protein mass is determined by adding the total mass of all the amino acids in the protein to the mass of any molecules added through post-translational modification. A protein's isoelectric point is the pH at which it is neutrally charged. A protein's N- and C-terminal sequence tags are short sequences of amino acids on either end of the protein. Since there are twenty different possible amino acids at each position in a protein, a **peptide** of only four or five amino acids in length is likely to be unique to a specific protein. There are 160,000 (20^4) combinations of sequences that are four amino acids long.

The most commonly used laboratory techniques in proteomics are two-dimensional polyacrylamide gel electrophoresis (2-D PAGE) and mass spectrometry. These techniques have been modified for use in proteomics. Both can be used in combination with more traditional protein separation techniques, including column chromatography.

peptide amino acid chain



Starting in the late 1990s, several companies also started developing “protein chips,” another strategy for studying proteomes and other complex protein mixtures. These chips allow a researcher to collect minute quantities of proteins that bind to specific molecules on their surface. By 2001, some companies announced they were developing “antibody chips” onto which antibodies will be attached. The antibodies can then be used as probes to capture and quantify specific proteins found in complex mixtures.

The use of 2-D PAGE allows the simultaneous separation of thousands of proteins, and the technique is still a key tool in proteomics technologies. The first dimension of protein separation on the gel is by isoelectric focusing, in which proteins are separated along a pH gradient until they reach a stationary position, where their net charge is zero.

The second dimension of separation on the gel is by molecular mass. Sodium dodecyl sulphate (SDS) is applied, and it binds to all the proteins. This provides the proteins with a uniform charge along their length, so that they will migrate across the gel according to their molecular mass when a current is applied. After the 2-D PAGE is run, the gel is stained. The result is a two-dimensional map consisting of hundreds or thousands of protein spots.

Since the early use of 2-D PAGE in the early 1970s, a number of modifications have been made to make gels more reproducible and more amenable to the higher-throughput use necessary for proteomics applications. However, 2-D PAGE is still something of an art form, and high-quality, reproducible results are difficult to obtain except in the hands of very experienced users. The technology needs to be further simplified to allow casual and novice users to obtain reproducible, quality results.

Mass spectrometry is an analytical technique that very accurately measures the mass of proteins and **peptides**. There are two common types of mass spectrometry. The first type, matrix-assisted laser desorption/ionization time-of-flight mass spectrometry, can be used to analyze proteins that are embedded in solid samples and measures their mass in a flight tube. The second type, electrospray ionization mass spectrometry, can be used to analyze proteins that are in a liquid solution and measures their mass in either a flight tube or in a device known as a quadrupole. There are also other variations on these techniques.

Mass spectrometry is commonly used for peptide mass fingerprinting. In this process, a protein sample is isolated by 2-D PAGE and cut with an enzyme that specifically targets particular amino acids. Mass spectrometry is used to measure the masses of the resulting cut pieces, or peptides. These masses can be thought of as a fingerprint that can be compared to the fingerprints of proteins whose amino acid sequences have already been analyzed and stored in a database.

To determine the fingerprints of proteins that have already been sequenced, a computer program determines the amino acid composition, and thus the masses, of the pieces that would result if those proteins were also cut by the same enzyme. A list of proteins is generated from the database, sorted by how many peptides they share with the unknown experimental protein.

peptides amino acid
chains

There are also technologies, including the yeast two-hybrid system, that can be used to study interactions between proteins. These approaches complement 2-D PAGE and mass spectrometry data by helping to elucidate functional cellular pathways.

Databases and Computational Approaches

There is an ever-increasing number of protein and proteome databases being developed. The most comprehensive information about specific proteins is found in databases that store protein sequences. One of the first and probably the best known such database is SWISS-PROT, which was created in 1986.

SWISS-PROT is a curated database that provides not only protein sequences but also such information as descriptions of a protein's function, its domain structure, and post-translational modifications, as well as links to other related databases. Other sequence-based protein databases include the Yeast Proteome Database and Human PSD.

There are also a number of widely used pattern and profile databases that are used to reveal relationships among proteins based on the presence of particular groups of amino acids in the proteins' sequences. Such groups, known as patterns, motifs, domains, signatures, or fingerprints, are found in specific regions of proteins that are important to some function of the protein. They could be in an area that performs some type of enzymatic activity or that is the site of a certain post-translational modification. Both their sequence and structure are typically well conserved. Some of the best known pattern and profile databases are: PROSITE, Pfam, PRINTS, and BLOCKS. SEE ALSO ALTERNATIVE SPLICING; BIOINFORMATICS; GEL ELECTROPHORESIS; GENOME; HUMAN GENOME PROJECT; MASS SPECTROMETRY; POST-TRANSLATIONAL CONTROL; PROTEINS.

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Pseudogenes

Pseudogenes are defective copies of functional genes. These may be partial or complete duplicates derived from polypeptide-encoding genes or RNA genes. The DNA sequence of a pseudogene is characteristically very similar to its functional counterpart, but contains variant mutations that render the gene inactive. The functional polypeptide-encoding gene contains an open reading frame, a long stretch of nucleotides that are transcribed and subsequently translated into a series of amino acids uninterrupted by stop **codons**. In contrast, pseudogenes derived from polypeptide sequences generally are punctuated with stop codons, effectively rendering them incapable of producing a functional protein.

Pseudogenes may also contain frameshift **mutations**, yielding a change in the reading frame. Additionally, there may be mutations that inactivate regulatory elements or intron-splicing sites. In either case, the duplicated

codons sequences of three mRNA nucleotides coding for one amino acid

mutations changes in DNA sequences



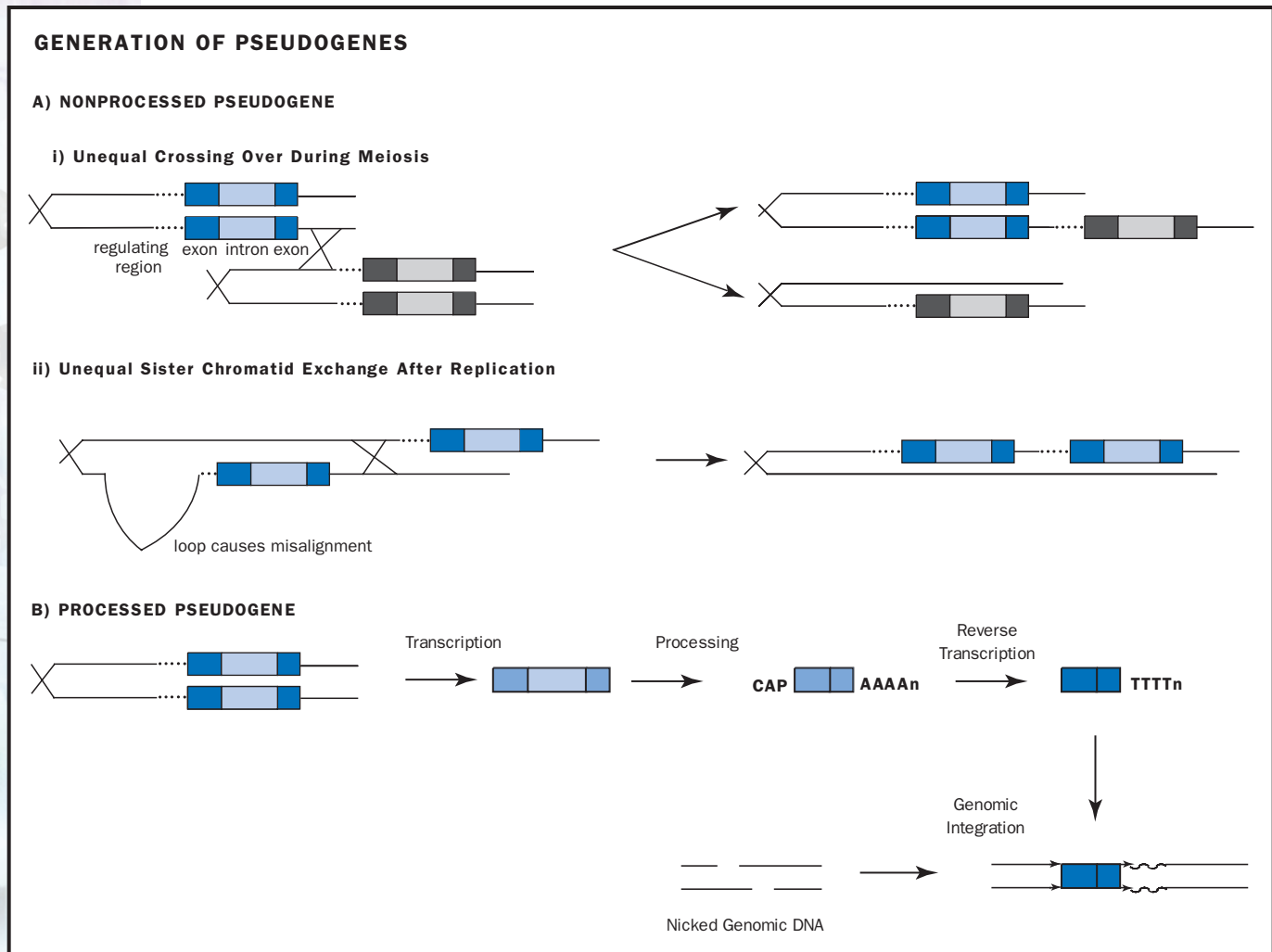


Figure 1. Generation of pseudogenes. (A) Direct duplication of a DNA sequence including regulatory elements and introns yields a nonprocessed pseudogene. (B) Duplication via an RNA intermediate, processed with a 3' polyadenylated tail, 5' cap (7-methylguanosine), and intron removal, yields a retropseudogene. The arrows indicate direct repeat sequences resulting from the integration. The A-T refers to a stretch of adenines and complement thymines in the newly derived pseudogene.

gene may be rendered nonfunctional. Genes and pseudogenes derived from the duplication of an ancestral gene are said to be **paralogous**.

Nonprocessed Pseudogenes

Gene duplication may occur by a direct increase of DNA content (non-processed) or via an RNA intermediate (processed). Nonprocessed pseudogenes can arise by unequal crossing over in homologous chromosomes (paired meiotic chromosomes containing the same genetic loci) or unequal sister **chromatid** exchange (crossover at improperly aligned sequences) after replication in a single chromosome (Figure 1A). Replication slippage also increases DNA content by looping of the synthesized strand during DNA replication, but typically involves short sequence stretches such as microsatellites (short repeated sequences).

A nonprocessed duplicated gene contains the introns and regulatory sequences of the original gene. This yields genetic redundancy, which allows

chromatid a replicated chromosome before separation from its copy

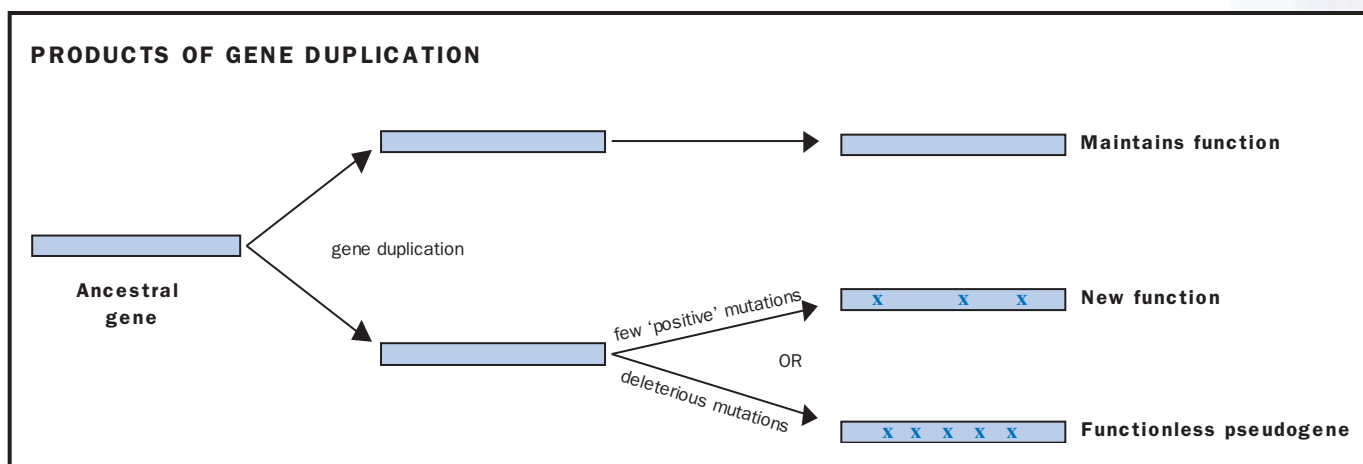


Figure 2. Gene duplication yields either the acquisition of a gene with novel function or a pseudogene. X refers to mutations.

one of the genes to acquire mutations, becoming a nonfunctional pseudogene (Figure 2). Occasionally, the duplicated gene acquires mutations yielding a gain-of-function that differs from the original gene (Figure 2). This may allow the evolution of new capabilities in the organism possessing it.

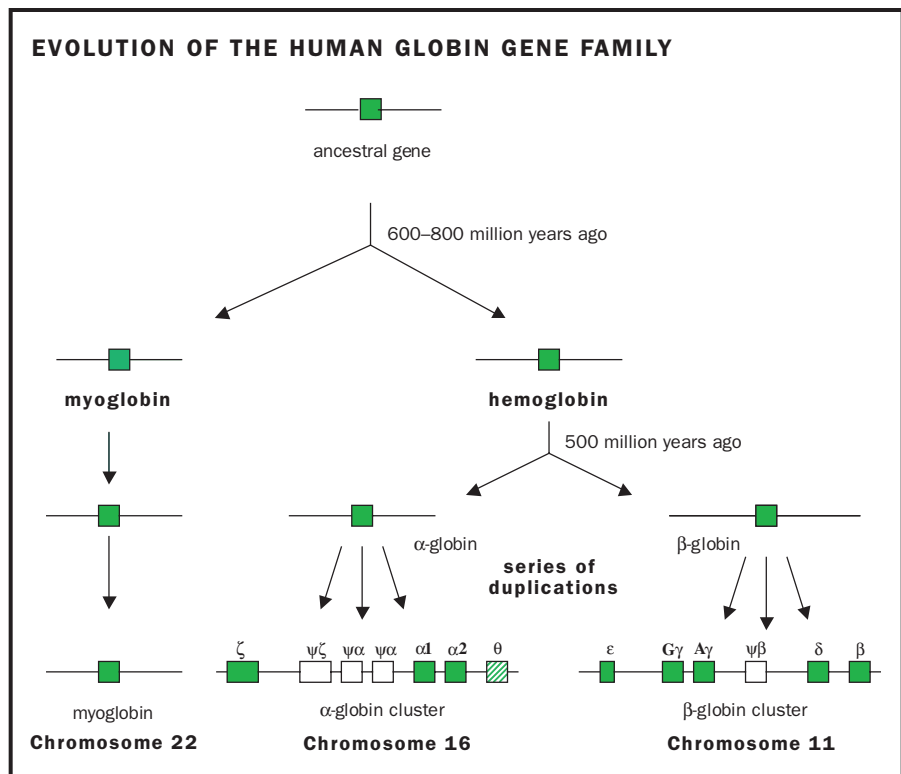
New genes generated by nonprocessed duplication are generally located in the vicinity of the ancestral (original) gene within the genome. However, it is possible that these genes may become separated from each other as a result of major chromosomal rearrangements (such as translocations). Examples of functional and nonfunctional duplicated genes in adjacent locations and on different chromosomes are exhibited by the globin gene superfamily.

Processed Pseudogenes

Pseudogenes generated via a messenger RNA (mRNA) intermediate demonstrate the features of processed RNA. These genes lack the flanking transcriptional regulatory sequences, do not contain introns, and typically have a polyadenylated 3' (3-prime) region (adenine-containing nucleotides are added to the mRNA in eukaryotes). These sequences are converted to complementary DNA (cDNA) by the enzyme reverse transcriptase, and then integrated back in the genome at a new location (Figure 1B). These elements, therefore, are not necessarily in the chromosomal vicinity of the original sequence, and are essentially dead on arrival. Processed pseudogenes are also referred to as retropseudogenes.

Processed pseudogenes may also be derived from other RNA genes, such as tRNA (transfer RNA), rRNA (ribosomal RNA), snRNA (small nuclear RNA), and 7SL RNA. Evidence for this phenomenon includes the identification of nonfunctional tRNA genes containing a CCA sequence at the 3' terminal. CCA is not part of the original DNA sequence, but is enzymatically added to the tRNA molecule following transcription. The gene for 7SL RNA is an integral component of the signal recognition particle complex involved in transmembrane protein transport. The 7SL RNA gene is thought to be the ancestral gene for the primate Alu and rodent B1 retroposons, based on sequence similarities. (Retroposons are a type of transposable genetic element that is found littered throughout the genome.) The *Alu* element differs from 7SL by having two internal sequence deletions,

Figure 3. Evolution of the human globin gene superfamily. Solid boxes represent functional genes, striped boxes represent expressed pseudogenes, and empty boxes represent pseudogenes.



duplication of the entire sequence, and numerous nucleotide substitutions. At some point in its evolutionary past a 7SL retropseudogene apparently integrated into a highly fortuitous location, as there are about 1.5 million *Alu* elements in the human genome, accounting for approximately 10 percent of our DNA. Most *Alu* elements are retropositionally incompetent pseudogenes, hence incapable of generating additional copies. Other retroposons are thought to be derived from processed duplicated tRNA genes (for example, rodent B2 and ID elements).

Pseudogene Examples

The globin gene superfamily provides an interesting example of the generation of both functional and nonfunctional duplicated genes (Figure 3). Based on nucleotide sequence data, it appears that a gene duplication occurred about 600 to 800 million years ago, yielding myoglobin and hemoglobin genes. Another duplication of the hemoglobin gene occurred about 500 million years ago, yielding α -globin and β -globin genes. (Adult human hemoglobin contains two α and two β strands.) These are all functional genes, found on three different human chromosomes. The α -globin and β -globin genes further duplicated, yielding both pseudogenes and functional genes. Possession of more than one globin gene provides a selective advantage because it compensates for the variation of oxygen in the prenatal versus postnatal environment.

The α -globin gene cluster consists of three functional genes and three pseudogenes. There is also an additional gene that is expressed but not incorporated into a hemoglobin molecule. In other words, this would be an example of an expressed pseudogene. The β -globin gene complex con-

α the Greek letter alpha

β the Greek letter beta

sists of five functional genes and one pseudogene. Examples of other processed polypeptide-encoding pseudogenes include those derived from actin, ferritin, and glyceraldehyde 3-phosphate dehydrogenase genes. SEE ALSO EVOLUTION OF GENES; GENE; GENE FAMILIES; HEMOGLOBINOPATHIES; READING FRAME; REPLICATION; RNA; RNA PROCESSING.

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Psychiatric Disorders

Genetic studies of psychiatric disorders have become an important specialty area within medical genetics. Much of the progress in the area is the result of advances in molecular genetics techniques, The Human Genome Project, developments in the neurosciences, and recent genetic findings in complex brain disorders such as Alzheimer's and Huntington's disease.

Psychiatric Disorders with Genetic Involvement

It is well accepted that many of the psychiatric disorders listed in the *Diagnostic and Statistical Manual*, fourth edition, have hereditary predispositions. (The *DSM-IV* is the official diagnostic manual for mental disorders in the United States.) In the last two decades, clinical genetic (family, twin, and adoption) studies have provided some valuable information about both the genetic and environmental aspects that lead to the development of these psychiatric disorders. More recently, a growing number of molecular genetics studies have provided more data about the genetics of the disorders. However, there has been little success in the identification of specific genes involved in the **etiology** of these complex disorders.

The *DSM-IV* includes more than 350 diagnoses and at least one third of the disorders have had one or more clinical genetic study completed. Additionally, some of the disorders have been subjected to molecular genetic studies. Several of the most frequently examined *DSM-IV* psychiatric disorders from a "genetic study" perspective are listed below:

- Alcoholism
- Alzheimers
- Attention Deficit Hyperactivity Disorder (ADHD)
- Autism
- Bipolar Disorder
- Mental Retardation
- Obsessive-Compulsive Disorder (OCD)
- Panic Disorder
- Schizophrenia

etiology causation of disease, or the study of causation

- Social Phobia
- Substance Use Disorder
- Tourette's Syndrome

Major psychiatric disorders, such as schizophrenia and bipolar disorder, have been shown in numerous studies to have significant genetic factors involved in their etiology. The effects of environment and gene-environment interactions on the expression of the disorders are thought to be important, but need more study. Several childhood onset disorders, such as mental retardation, autism, and ADHD, also have strong genetic components in their etiology. All of these disorders are considered complex and are thought to have multiple genes acting together with nongenetic factors, with each gene typically contributing only small effects.

Inheritance Patterns and Linkage Studies

Complex disorders such as schizophrenia and bipolar disorder usually do not follow classic Mendelian inheritance patterns, but they can frequently mimic a pattern of autosomal dominance with reduced penetrance. Explanatory models for these complex disorders include multifactorial inheritance (multiple genes with nongenetic components) and epistasis (few genes acting jointly). But, since the mode of inheritance is unknown, a range of analytic methods must be used to study the genetic aspects of these disorders. Linkage and allelic association studies are frequently methods used to investigate possible causal genes for complex psychiatric disorders. However, in large populations, there are likely to be several causal or susceptibility genes and nongenetic causes, as well.

Historically, the first positive linkage study to have a major impact on psychiatry was the linkage of bipolar disorder to chromosome eleven in a large Amish family. However, a later assessment of the family failed to confirm the original linkage results. A similar situation occurred with the failure to replicate the findings of an early report of linkage of schizophrenia to a region on chromosome 5 in Icelandic and British families.

As a result of these early problems, it became clear that more rigorous methodology, such as more careful clinical phenotyping, the use of more genetic markers, and new analytic techniques would be necessary if molecular approaches were to be used in finding genes involved in psychiatric disorders.

Clinical genetic (family, twin, and adoptive) studies have been attempted in a large number of psychiatric disorders. However, these studies generally provide no information about what genes are involved in the disorder. Molecular studies are necessary to begin to elucidate this data. Schizophrenia and bipolar disorder are examples of psychiatric disorders that have been studied for nearly two decades.

Schizophrenia and Bipolar Disorder

Schizophrenia is a disorder characterized by psychotic symptoms such as hallucinations, delusions, and disordered thinking, as well as deficits in emotional and social behavior. Numerous family, twin, and adoption studies have provided substantial evidence for genetic factors in the **etiology** of this dis-

etiology causation of disease, or the study of causation

order. Nongenetic factors also appear to play an important role. The risk for relatives of an individual with schizophrenia is 6.5 percent, nearly eight times the population rate of the disorder. The concordance rate for monozygotic (identical) twins is 45 percent, and for dizygotic (fraternal) twins the rate is 12 percent. The higher rate for monozygotic twins, who share all their genes, is strong evidence that genes play a role in schizophrenia.

Molecular genetic (linkage and association) studies suggest that there are multiple genes with small effects that predispose one to the disorder. In an early study, chromosome 5q was implicated, but was not replicated. Areas on chromosome 6p and 8p have been replicated in at least one follow-up study. More recently, chromosome 1 has been implicated. While no causative genes have been identified, there have been molecular studies that implicate chromosomes 1q, 3p, 5, 6p, 6q, 8p, 10p, 13q, 18p and 22q in schizophrenia. Like other multifactorial diseases, the etiology of schizophrenia involves multiple genes and gene-environment interactions.

Bipolar disorder is characterized by episodes of depression and mania, elevated or irritable mood, and symptoms such as rapid thoughts, grandiose ideas, and reckless behavior. Population, twin, and adoption studies provide evidence for the role of genetics in the etiology of bipolar disorder. First-degree relatives have about a 7 to 10 percent risk of having the disorder once one family member is diagnosed. The concordance rate for monozygotic twins is 60 to 65 percent, and for dizygotic twins the rate is 10 to 15 percent, the same as for non-twin siblings.

One of the first bipolar disorder molecular genetic studies implicated chromosome 11, but this finding was not replicated in several other studies. A similar failure to replicate occurred with the initial reports of linkage on the chromosome X. Regions on chromosome four are reported to show strong evidence of linkage to some bipolar families. Major efforts in the last few years have been focused on chromosome seven and eighteen. Some of the studies have suggested a parent-of-origin effect, with maternal transmission more common than paternal. Both linkage and association studies have implicated chromosomes 4p, 6p, 12q, 13q, 16p, 18p, 18q, 21q, and 22q in bipolar disorder. Major depressive disorder has also been studied, but the underlying genetic factors have not been identified.

Tourette's Syndrome

Some psychiatric disorders, like Tourette's syndrome, have significant overlap (co-morbidity) with other psychiatric disorders (OCD and ADHD in the case of Tourette's). This may make the elucidation of genetic causes more difficult.

Tourette's syndrome is characterized by multiple motor tics and one or more vocal tics. It has onset before eighteen years and is one and one-half to three times more common in males. Twin, adoption, and segregation analysis studies support a genetic etiology for Tourette's syndrome. An **autosomal** dominant inheritance was initially suggested, but other inheritance patterns have been recently reported. There appears to be a relation with ADHD (up to 50 percent of individuals with Tourette's also have ADHD) and OCD (up to 40 percent of individuals with Tourette's have OCD). First-degree relatives are at high risk for developing tics and obsessive compulsive disorders.

autosomal describes a chromosome other than the X and Y sex-determining chromosomes



Studying psychiatric disorders for genetic factors in their etiology is difficult because of the phenotype definition, co-morbidity, and multiple causal factors. The summary finding from the genetic studies to date suggest that there are multiple genes of small to moderate effect underlying the predisposition to these complex psychiatric disorders. SEE ALSO ALZHEIMER'S DISEASE; ATTENTION DEFICIT HYPERACTIVITY DISORDER; COMPLEX TRAITS; GENE DISCOVERY; INHERITANCE PATTERNS.

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Public Health, Genetic Techniques in

As of 2002 more than ten thousand genes have been discovered, and it is estimated that 30,000 to 70,000 human genes will be identified as a result of the Human Genome Project over the following few years. Tests for more than 600 gene variants are already available in medical practice.

Genetic variants, or polymorphisms, are a normal part of genetic variability that may or may not be associated with an increase or decrease in disease risk. With advances in biotechnology, newly characterized genetic variants are being identified at a rapid rate. The challenge will be to ensure the appropriate use of genetic information to improve health and prevent disease in individuals, families, and communities.

The broad mission of public health is to act in society's interest to assure conditions in which people can be healthy. Public health genetics is the application of advances in genetics and molecular **biotechnology** to improve the public's health and prevent disease. Rapid progress in biotechnology, in sequencing the human genome, and in characterizing gene expression have generated high hopes of finding new ways to improve the public's health through the prevention and treatment of diseases.

At the same time, concerns have been raised about the premature application of such technologies and knowledge. Genetic testing and the understanding of how factors such as **hormones**, diet, and the environment influence risk in susceptible individuals and human populations present important challenges in the field of public health genetics. Ethical issues, health care policy priorities, risks of discrimination in employment and insurance, and complex psychological aspects within families are also crucial issues in this field.

Public Health Approaches in Genetics

Disease-related genetic variants that have been characterized include those associated with rare diseases as well as those that increase susceptibility to

biotechnology production of useful products

hormones molecules released by one cell to influence another

common chronic diseases such as cancer and heart disease. Risk for almost all human diseases results from the interactions between inherited gene variants and environmental factors, including chemical, physical, and infectious agents, as well as behavioral or nutritional factors. Thus it appears reasonable to direct disease-prevention and health-promotion efforts toward individuals at high risk because of their genetic makeup.

To function effectively, public health genetics needs to meet several challenges, including (1) the implementation of research, (2) the evaluation of genetic information and tests, (3) the development, implementation, and evaluation of population interventions, and (4) effective communication and information dissemination. With the realization that all human disease is the result of interactions between genetic variation and the environment (dietary, infectious, chemical, physical, and social factors), it becomes evident that it is important to identify the modifiable risk factors for disease that interact with the genetic variation. This approach can be used to help develop preventive strategies. To be able to deliver appropriate genetic tests and services for disease prevention and health promotion, it is important to integrate genetic services into disease-prevention and health-promotion activities.

Applied Research

Public-health assessment relies on scientific approaches such as surveillance and **epidemiology** to assess the impact of discovered gene variants on the health of communities. Surveillance, which involves the systematic gathering, analysis, and dissemination of population data, is needed to determine population frequencies of genetic variants that predispose people to specific diseases. This information can be used to assess the population morbidity and mortality associated with such diseases. Surveillance can also be used to determine the prevalence and effect of environmental factors known to interact with given genotypes. Additionally, surveillance can aid in the evaluation of a genetic test in terms of its safety, effectiveness, and cost-effectiveness.

Another example where surveillance can come into play in public health genetics is with infectious diseases and DNA fingerprinting. The ability to characterize **pathogenic** organisms phenotypically and genotypically can be a powerful approach that provides information important for an antimicrobial surveillance program. It also is a means of providing information that may be useful for understanding pathogenic microorganisms worldwide. With such a program in place, the clonal spread of multiresistant pathogens among patients can be identified.

Epidemiology is the study of the distribution and determinants of health-related states or events in populations and is used to investigate risk factors for various diseases and identify high-risk populations. Epidemiological information can be used to target populations that could benefit most from prevention and intervention actions. Epidemiology is also a tool for evaluating the effect of health programs and services on a population's health. Population-based epidemiological studies are increasingly needed to quantify the impact of gene variants on the risk of disease, death, and disability, and to identify and quantify the impact of modifiable risk factors that interact with gene variants.

epidemiology study of incidence and spread of diseases in a population

pathogenic disease-causing



metabolites molecules involved in metabolic pathways

Evaluation of Genetic Information and Tests

Genetic tests include the analysis of human DNA, RNA, chromosomes, proteins, and certain **metabolites** to detect a person's genotype for clinical purposes, including predicting the risk of disease, identifying gene mutation carriers, and establishing prenatal and clinical diagnoses or prognoses. Successful implementation of genetic tests to improve public health requires careful assessment of how and when genetic tests can and should be used to promote health and diagnose and prevent disease. This assessment must include the development of standards and guidelines for assuring quality genetic testing, and the consideration of ethical and legal issues.

Genetic tests need to be evaluated on the basis of several parameters before they can be taken from research laboratory to clinic. It is necessary to assess (1) how good the test is in predicting the underlying genotype, (2) how good the test is in diagnosing or predicting the phenotype or disease, and (3) the benefits and risks of the genetic test and ensuing interventions. Genetic test validity is quantified in terms of sensitivity (the probability of testing positive for the genetic test if there is a gene mutation or if disease occurs), specificity (the probability of testing negative if the genetic mutation is not present or if the disease does not occur), and predictive value (the test's ability to accurately predict disease).

All clinical laboratories in the United States that provide information to referring physicians are certified under the Clinical Laboratory Improvement Act (CLIA) amendments of 1988. The CLIA standards for quality control, proficiency testing, personnel, and other quality assurance practices apply to all genetic tests.

Development, Implementation, and Evaluation of Population Interventions

Recent advances in human genetics have brought high expectations for implementing prevention strategies among genetically susceptible individuals. Yet the clinical use of this information poses risks as well. It is the role of public health to develop intervention strategies for diseases with a genetic component, implement pilot demonstration programs, and evaluate the impact of the intervention on reducing morbidity and mortality in the population. This evaluation includes conducting a needs assessment of genetic services, studying the impact of genetic counseling on public health, and applying prevention-effectiveness principles to genetics programs. Policy analysis of **informed consent** to genetic testing, stigmatization of individuals and groups, discrimination in employment, and access to insurance need to be considered.

The two recently identified susceptibility genes *BRCA1* and *BRCA2*, which are associated with a high risk of developing breast and ovarian cancers, illustrate some of the complexities individuals from high-risk families face. Studies to determine the efficacy of prophylactic surgeries, chemoprophylactic strategies, and other preventive measures have not been conclusive, leaving individuals from high-risk families—and those who are carriers of mutations—with complex decisions concerning genetic testing and medical intervention.

Newborn screening illustrates the evolution of effective genetic-screening programs. Recognizing the potential importance of phenylketonuria

informed consent knowledge of risks involved

screening over forty years ago, the Children's Bureau (now called the Maternal and Child Health Bureau) sponsored a multistate urine-screening program. The initial outcome was that 30 percent of infants remained untested due to inadequate specimen-collection and delivery. There was sometimes a false-positive reading. Improvements were made, and today a single drop of blood, rather than a urine sample, is sufficient for eight to ten assays for metabolic-disease indicators, along with genetic and infectious information about the mother.

Another example of an evolving genetic-screening program involves the common abnormal hemoglobin "S," or sickle hemoglobin, detected in newborn-screening programs in the United States. This hemoglobin is the defining characteristic for sickle cell disease. The first statewide screening program was established in 1975. Widespread acceptance and implementation was lacking until after a 1986 study showed the efficacy of daily oral penicillin prophylaxis in preventing infection among young children with sickle cell anemia. The efficacy of sickle cell screening was demonstrated through epidemiological efforts to evaluate pediatric outcomes after newborn screening, by demonstrating that mortality rates declined from 1968 to 1992, particularly in cohorts of sickle cell patients.

Communication and Information Dissemination

The fourth public-health function in genetics is developing and applying communication principles and strategies related to advances in human genetics, interventions, and genetic tests and services, as well as in interventions, the ethical, legal, and social issues related to these topics. Public health agencies can play a role in translating the very complex information related to genetics and disease prevention to health care workers and the public. An appropriate mix of mechanisms should be used to disseminate information, including distance-based interactive meetings, information centers, and electronic communication.

Summary

In summary, four public health functions for genetics have been outlined that underscore the complexities involved in public health genetics. All are carried out by efforts among various groups, including partnerships and coordinated efforts among federal, state, and local agencies, the public and private sectors, and the public-health, medical, and academic sectors, with various levels of community and consumer involvement. Annual national meetings on genetics and public health facilitate these efforts. **SEE ALSO** ANTIBIOTIC RESISTANCE; CANCER; DIABETES; DISEASE, GENETICS OF; GENETIC DISCRIMINATION; GENETIC TESTING; HEMOGLOBINOPATHIES; METABOLIC DISEASE; PRENATAL DIAGNOSIS.

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Purification of DNA

Many procedures in molecular biology require an initial pure sample of DNA. These procedures include the polymerase chain reaction, sequencing, gene cloning, blotting, and DNA profiling. Purification of DNA involves removing it and other constituents from the cell, separating it from the various other cell constituents, and protecting it from degradation by cellular **enzymes**. Isolation procedures must also be gentle enough that the long DNA strands are not sheared by mechanical stress.

DNA can be isolated from almost any cellular source. White blood cells and cheek cells taken directly from humans are most commonly used for diagnostic purposes, but skin, hair follicles, semen, and other tissues can be used for **forensic** analysis. Cells grown in petri dishes or in suspension can also be used. The cells are isolated from any surrounding fluid (such as blood serum) by centrifuging them—spinning them at high speeds—and then are resuspended in a buffer solution. The buffer prevents rapid or dramatic changes in pH, which can interfere with subsequent reactions. To break open the cell membranes, a detergent is added to the buffer. Sodium dodecyl sulfate (SDS) is often used for this purpose. The detergent also helps remove proteins and lipids in the cell.

enzymes proteins that control a reaction in a cell

forensic related to legal proceedings

The buffer also contains ethylenediaminetetraacetic acid (EDTA), which is a chelator. Chelators are molecules that act as scavengers for metal ions in solution. This is important because DNase, an enzyme that digests DNA, is present in the cell and would destroy the long DNA strands if it was active. DNase activity requires magnesium ions, and EDTA removes them from solution, preventing DNase from cutting up the DNA. RNase is also present in the buffer at this step, to break up the RNA present in the cells.

The solution is then treated with proteinase K, a highly effective enzyme that inactivates all types of proteins. This enzyme can also be used earlier in the procedure to break apart clumped cells. Unlike many proteins, proteinase K remains active at elevated temperatures, so the solution can be heated to about 55 °C to aid protein inactivation and removal by the detergent. This step may last between two and sixteen hours.

Once the cells are broken open and the RNA, proteins, and lipids have been dissolved in the buffer, the DNA must be separated from these materials. One standard technique uses phenol to remove the proteins, leaving DNA and other water-soluble materials behind. The DNA is then extracted from the water phase using chloroform and precipitated from the chloroform using ethyl alcohol mixed with sodium acetate salt. The DNA is then removed either by spooling the long threads onto a glass rod, or by spinning it out of solution using a centrifuge. The DNA is then resuspended in **buffer**.

Another technique for separating the DNA from the mixture avoids the use of phenol and chloroform, which are toxic. Instead, the proteins are “salted out” by adding a concentrated salt solution and then removed by centrifuging. In another method, DNA is adsorbed onto very small glass beads, in the presence of “chaotropic” salts, which disrupt protein structure. The beads are removed (or washed in place), and the DNA is released by changing the salt concentration.

Once a pure sample of DNA has been obtained, the fragments it contains may be separated on the basis of size by **gel electrophoresis**. Specific sequences can be identified by **Southern blotting** using probes with complementary sequences, and they can then be cut out of the gel for further use, such as cloning. If the sample is small, the DNA can be amplified by the **polymerase** chain reaction.

A crude preparation of DNA can be made in the kitchen with simple ingredients, including baking soda (which acts as a buffer), laundry detergent, table salt, and rubbing alcohol. Meat tenderizer (an enzyme preparation) can be used to destroy proteins. **SEE ALSO** **BLOTTING**; **CLONING GENES**; **DNA**; **DNA PROFILING**; **GEL ELECTROPHORESIS**; **POLYMERASE CHAIN REACTION**; **SEQUENCING DNA**.

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buffers substances that counteract rapid or wide pH changes in a solution

gel electrophoresis technique for separation of molecules based on size and charge

Southern blotting separating DNA fragments by electrophoresis and then identifying a target fragment with a DNA probe

polymerase enzyme complex that synthesizes DNA or RNA from individual nucleotides

